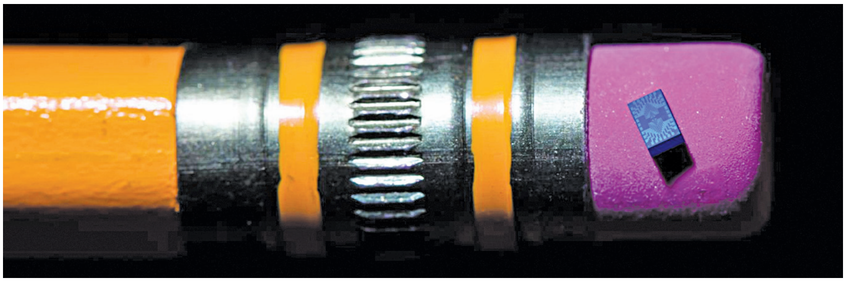
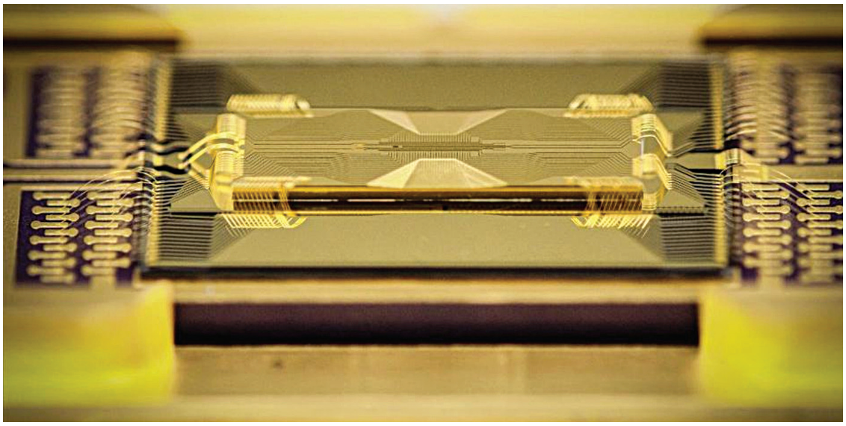
When asked why Intel focuses on silicon qubits in an IEEE Spectrum interview, Jim Clarke, director of Quantum Hardware at Intel said, "Silicon spin qubits look exactly like a transistor... The infrastructure is there from a tool-fabrication perspective. We know how to make these transistors. So, if you can take a technology like quantum computing and map it to such a ubiquitous technology, then the prospect for developing a quantum computer is much clearer."

Trapped ion based: In this system, the qubits are ions trapped by electric fields and controlled by lasers, as the name suggests. The qubits are long-lived because trapped ions have relatively long coherence periods. They can also communicate with each other without difficulty.

Trapped-ion qubits are the technology of choice for research teams at numerous universities across the world as well as at firms doing similar work, like Honeywell and IonQ. In fact, IonQ has released what they claim to be the industry's first programmable multicore quantum architecture (RMQA), which is based on trapped-ion qubits.

Infineon Technologies and Oxford Ionics announced a partnership in July 2022 to develop highly effective, fully integrated QPUs. The innovative electronic qubit control (EQC) technology from Oxford Ionics provides a way to incorporate trapped ion qubits into Infineon's established semiconductor manufacturing processes.


Intel's silicon spin qubit chip balanced on a pencil eraser (Credit: Intel)


IonQ's EGT series ion trap chip (Credit: IonQ)

Superconductor based: One of the most promising methods for creating quantum computers is using superconducting qubits. Superconductivity, being a quantum phenomenon, permits us to employ superconducting circuits as controllable quantum systems. The Eagle quantum processor, based on the transmon superconducting qubit architecture, was introduced by IBM Quantum in November last year. It has 127 qubits. By 2023, the company plans to release a processor with 1121 qubits.

How to select a suitable QPU for your application

Every firm has a different set of insights that it needs in order to make the best choices. Based on your organisation, your current applications, and your computational needs, you need to take a decision on whether to use QPUs or not. First, you need to check if you have truly potential applications of quantum computing, and whether you have the funds to facilitate them.

As seen previously in this article, every quantum computer is built using a somewhat distinct hardware design and a multitude of qubit kinds. Each can also have a different architecture. Because

Some interesting developments in the quantum world

- In June 2022, Rigetti Computing, a pioneer in full-stack quantum-classical computing, announced the launch of its 32-qubit Aspen-series quantum computer in the UK.
- Nvidia, in July 2022, announced its new Quantum Optimized Device Architecture (QODA) framework, which will help accelerate developers' efforts around hybrid classical-quantum projects.
- For 2022 and 2023, IBM is developing processors with 433 and 1121 qubits, respectively.
- Google plans to develop a 1-million qubit processor by 2029.
- In April 2022, researchers at Intel and QuTech created the first silicon qubits at scale at Intel's D1 manufacturing factory in Hillsboro, Oregon.
- Recently, a QPU developed by Canada based company Xanadu has outrageously outperformed a classical system in a computing task, bringing down 9000 years of compute to 36 microseconds!